

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

CS XI

Chapter 6: Digital Literacy

Lecture 13 of 17

★ Golden

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection ★ Golden

Step 4: Prepare the survey

LEARN IT

Design **clear, unbiased** questions. Each question is linked to the type of data it produces:

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Step 5: Choose the sample

LEARN IT

The **sample** is the group of people who provide the data. It is usually impossible to ask everyone (the whole **population**), so a smaller group is chosen. The students decide to survey **40 Class XI students from different sections** to get a **representative** mix of habits. "Representative" means the sample looks like the whole group – it includes different kinds of students, not just your friends.

Step 6: Collect primary data (ethically)

LEARN IT

Ethics means doing what is morally right. When conducting the survey, students must follow strict ethical guidelines:

- Ask **permission** before collecting responses (**consent**).
- Avoid collecting **sensitive personal information**.
- Do **not record names** unless absolutely necessary (**anonymity**).
- Keep the responses **private**.
- Use the data **only for educational purposes**.

WORKED EXAMPLE

Why "different sections" matters. Suppose the group surveyed only the 40 students of the Pre-Engineering section who sit in the front rows.

Problem: these students might study more than average, so the results would show less phone distraction than is true for the whole class.

Fix: pick students from every section (e.g. 10 each from Pre-Medical, Pre-Engineering, Computer Science and Commerce). Now the sample better represents all Class XI students.

Ethical opening text for the form: "This survey is for a class project on phone use and study habits. It is anonymous – do not write your name. Your answers will be kept private and used only for this project. Do you agree to take part? ☐ Yes ☐ No"

REMEMBER IT

Ethics in five words: **Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE"**. Think: "Cover the person's nape (neck) – protect them."

COMMON MISTAKE

Mistake: Thinking a bigger sample is always enough. A sample of 200 students all from one section is still biased. **Fix:** A sample must be **representative**, not just large.

اردو میں سمجھیں

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔ پانچویں قدم میں نمونہ (Sample) چنا جاتا ہے: مختلف سیکشنز کے چالیس طلبہ تاکہ نتائج پوری جماعت کی نمائندگی کریں۔ چھٹے قدم میں اخلاقی اصولوں کے ساتھ ڈیٹا جمع کیا جاتا ہے: اجازت لیں، حساس معلومات نہ لیں، نام نہ لکھیں (گمنامی)، جوابات کو راز میں رکھیں اور صرف تعلیمی مقصد کے لیے استعمال کریں۔

EXAM TIP

Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words **consent** and **anonymity** – examiners look for them. You may also be asked to classify the four survey questions by data type.

CHECK YOURSELF

1. What is a sample?
2. What does anonymity mean in a survey?
3. What type of data does "Suggest one way to use phones responsibly" give?

Answers: 1. The group of people who provide the data. 2. Not recording the names of participants (unless absolutely necessary). 3. Qualitative.

Next lecture: 6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation